JAYAKAR COMMITTEE RECOMMENDATIONS (1927)

(With Reference to Ongoing Road Projects in India)

1. Introduction

The Jayakar Committee was appointed in 1927 to study road development in India. It laid the foundation for modern highway planning and recommended a systematic, scientific, and centralized approach for road construction and maintenance.

2. Recommendations on Ongoing Projects

2.1 Systematic Planning of Roads

- Road projects should follow a planned network system.
- Priority should be given to economic and strategic importance.

Example: A highway connecting industrial areas to ports (e.g., Mumbai–Pune Expressway).

2.2 Central Road Authority

- Establish a central authority to coordinate road projects across states.

Outcome Today: Indian Roads Congress (IRC) and MoRTH.

2.3 Dedicated Road Fund

- Create a Road Development Fund using fuel taxes.

Example: Central Road and Infrastructure Fund (CRIF).

3. Materials Used in Road Construction

3.1 Recommended Materials

- Soil (subgrade)
- Aggregates (coarse and fine)
- Bitumen (flexible pavements)
- Cement concrete (rigid pavements)
- Water (for compaction and mixing)

3.2 Modern Application

- Bituminous concrete for highways
- Cement concrete roads in urban areas
- Geosynthetics for soil stabilization

Example: Delhi–Meerut Expressway uses high-quality materials.

4. Type of Roads

- National Highways (NH)
- State Highways (SH)
- Major District Roads (MDR)

- Other District Roads (ODR)
- Village Roads (VR)

Example: NH-44 connects Srinagar to Kanyakumari.

5. Design Aspects

5.1 Geometric Design

- Proper alignment
- Safe curves and gradients
- Adequate sight distance

5.2 Pavement Design

- Strong subgrade
- Sub-base, base course, and surface course layers

5.3 Traffic Consideration

- Design based on traffic volume and load
- Provision for future expansion

Example: 4-lane highways expandable to 6 lanes.

6. Key Guidelines

- Scientific road development
- Proper maintenance system
- Classification-based planning
- Dedicated funding mechanism
- Research and training institutions
- Coordination between agencies

7. Output / Impact

- Formation of Indian Roads Congress (IRC)
- Establishment of Central Road Fund
- Development of Nagpur Road Plan (1943)
- Modern highway engineering practices

8. Conclusion

The Jayakar Committee transformed India’s road infrastructure through planned development, proper material usage, scientific design, and structured guidelines. Its recommendations remain the backbone of modern highway projects.

9. Example (Real-Time Application)

- Project: Delhi–Mumbai Expressway
- Type: National Highway
- Materials: Bitumen, concrete, aggregates

- Design: 8-lane expandable highway
- Guidelines: scientific planning, funding, advanced design